# Experiment 28
# Lenses

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**Abstract**

This experiment investigates the image formation through lenses, specifically converging (convex) and diverging (concave) lenses. The primary objective was to experimentally determine the focal lengths of these lenses and, from them, compute their optical powers. The study utilizes the ray diagram approach, focusing on the key optical principles such as the Paraxial Optics and lens equations to analyze image positions. A convex lens was used to produce a real image, which then served as the object for the concave lens. The experimental setup was validated by comparing theoretical values with results derived from measurements. Additionally, the experiment involves calculating uncertainties to provide a comprehensive error analysis of the measurements.

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# 1 Introduction

## 1.1 Contextualization

The study of the behavior of light is fundamental for understanding physical phenomenons and for the development of useful devices, for instance: cameras, telescopes and microscopes[1].

In this experiment we are going to analyze image formation through lenses, focusing on converging and diverging lenses, also known as convex and concave diopters respectively. The

main objective of this experience is to experimentally determine the focal lengths of these lenses and, from these, obtain their power [2].

### 1.2 Theorical principles

A lens is an optical device composed of a transparent material bordered by two curved (usually spherical) surfaces, whose main significance is its ability to form images of objects [1]. The qualities of these images depend on several elements, including the object's lighting, its dimensions, the material of the lens (specifically, its refractive index), and its physical design (such as its thickness and the curvature radii of its surfaces).

For the purposes of this experiment, we will treat the lenses as thin, meaning their thickness is small enough to be disregarded when compared to the radii of curvature of their outer surfaces [2].

The following image is a sketch, commonly known as ray diagram, that explains the behavior of the light rays when they go through a lens.

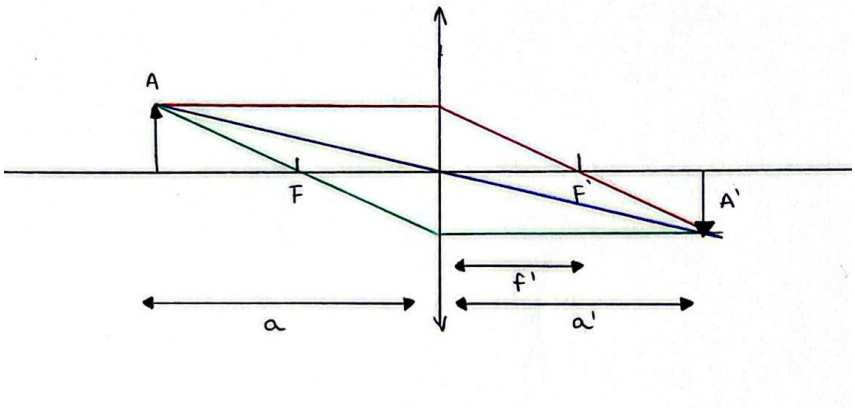



Figure 1: Ray diagram of a convex lens, where $a$ is the distance from the object to the lens, $a$' refers to the separation between the lens and the image it forms, and $f$' is the focal length.

The equation 1 is required to determine the position and size of the image of the object once it has gone through the lens.

$$\frac{1}{f'} = \frac{1}{a'} - \frac{1}{a} \tag{1}$$

where each length can be measured in $m$, $cm$, $mm$..., depending on what benefits us, and the result will be in the units we chose previously.

This expression holds true as long as the angles within the optical system remain small. In such situations, the considerable distance between the object and the lens—compared to the object's size—and the use of narrow apertures ensure this requirement is met. When these conditions apply, the system is said to operate within the realm of Paraxial Optics.

In Paraxial Optics the following rules must be taken into account.

- The light rays travel from left to right. Therefore, the object will be positioned at the left of the lens.
- The center of the coordinates system is set in the middle of the lens. This is the reason why the distances on the left on the lens are negative, whereas the ones on the right are always positive.

- Finally, objects are real when $a < 0$ and virtual if $a > 0$. On the contrary, when the image of the object forms on the left of the lens ($a' < 0$) it is virtual, and when it originates on the right ($a' > 0$) it is real.

In order to obtain the focal length of the lenses we need to clear $f$' from equation 1, resulting as:

$$f' = \frac{a \cdot a'}{a - a'}(m) \tag{2}$$

where $a$ and $a$' are measured experimentally.

The power of the lens os defined as the inverse of the focal length. It is measured in diopters ($m^{-1}$).

$$\psi[D] = \frac{1}{f'[m]} \tag{3}$$

The position and size of an image formed by a thin lens can be determined graphically in a quite straightforward manner. This method involves finding the intersection point of two rays originating from the same point on the object. In the context of Paraxial Optics, it is assumed that all rays from that point, regardless of their direction, will converge at the same location after passing through the lens, forming the image at that point. To be more exact we can use three rays instead of two.

1. A ray that travels parallel to the optical axis will go through the image focus $F$' after passing through the lens. This focus lies on the optical axis at a distance $f$' from the lens. In the case of a diverging lens, the ray does not pass through the image focus directly since it is located on the left of the lens in concave lenses. Instead, its extension does.

2. A beam passing through the object focus point F travels parallel to the optical axis after refracting in the lens.

3. A ray that goes through the center of the lens is not deviated.

With this method it is possible to obtain the position of the image in any situation. Figure 1 shows a ray diagram of a convex lens, whereas figure 2 represents the behaviour of light when refracting in a diverging lens.

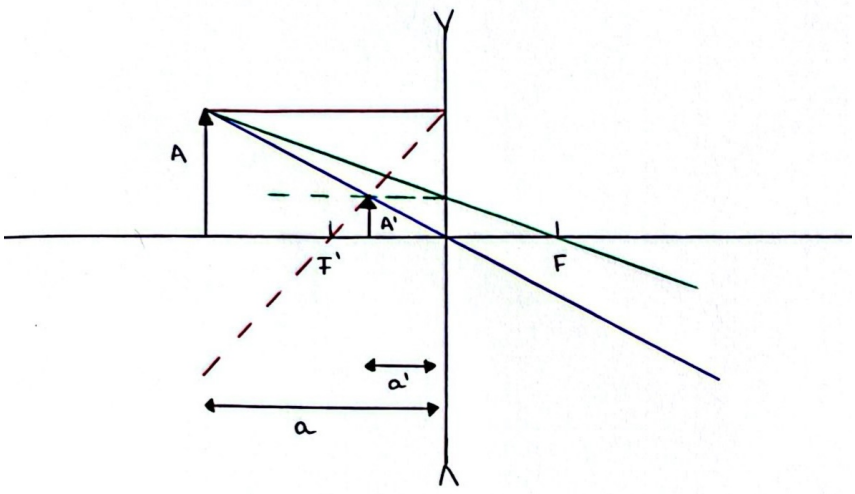



Figure 2: Ray diagram of a diverging lens

# 2 Materials and Methods

## 2.1 Materials

The materials that were used in this practice are the following[2]:

- **Optical bench**. With an illumination source, it will enable us to illuminate the object and see its image.
- **Sliding supports**. With them, we will move the object, lens or screen along the optical bench.
- **Screen**. We will collect the formed image there.
- **Convex lenses** whose focal length, actually, we know: $100 \pm 1$ mm, $150 \pm 1$ mm and $200 \pm 1$ mm.
- **Concave lens** whose focal length we also know: $-100 \pm 1$ mm.
- **Object**. In this case, as it must be something not symmetrical, it will be an arrow.
- **Ruler**. With a resolution of $0,001$ m, we will use it to determinate the other material's position.



Figure 3: Convex lens with a 150 mm focal length.



Figure 4: On the rigth, the convex lens and the screen.

## 2.2 Methods

### 2.2.1 Determination of the focal length of a convex lens

First, we must check that both the light source with the object (on the far left) and the screen (on the right) are correctly positioned on the optical bench. Once confirmed, the converging lens will be placed between these two elements and moved slowly along the bench until the image projected on the screen appears sharp. It is important to note that:

- If the object-lens distance is less than the focal length of the lens, the image will be virtual since the rays do not converge on the right side of the lens, making it impossible to project onto the screen.
- If the system is not properly aligned, as the screen is moved away from the object, the image will either not form or appear blurry.

Once a clear image is achieved, we will be able to take the positions of the object, the lens and the screen and, with their values, calculate necessary distances to find the focal length of the lens using equation 1.

Additionally, equation 3 will be applied to determine lenses' optical powers based on their obtained $f\prime$ value.

Finally, we will draw a ray-tracing diagram to verify if the results are consistent and confirm the correct application of the method, using the theoretical value of the lenses' focal lengths.

### 2.2.2 Determination of the focal length of the concave lens

In this part of the experiment, it is important to note that concave lenses will not be used alone, as they produce a virtual image that cannot be projected onto a screen.

The optical bench will be set up as in the previous section: with a convex lens forming a real image, which can indeed be captured on the screen. This image will act as a virtual object for the concave lens that we will now place in the set up, which will then produce a real image:

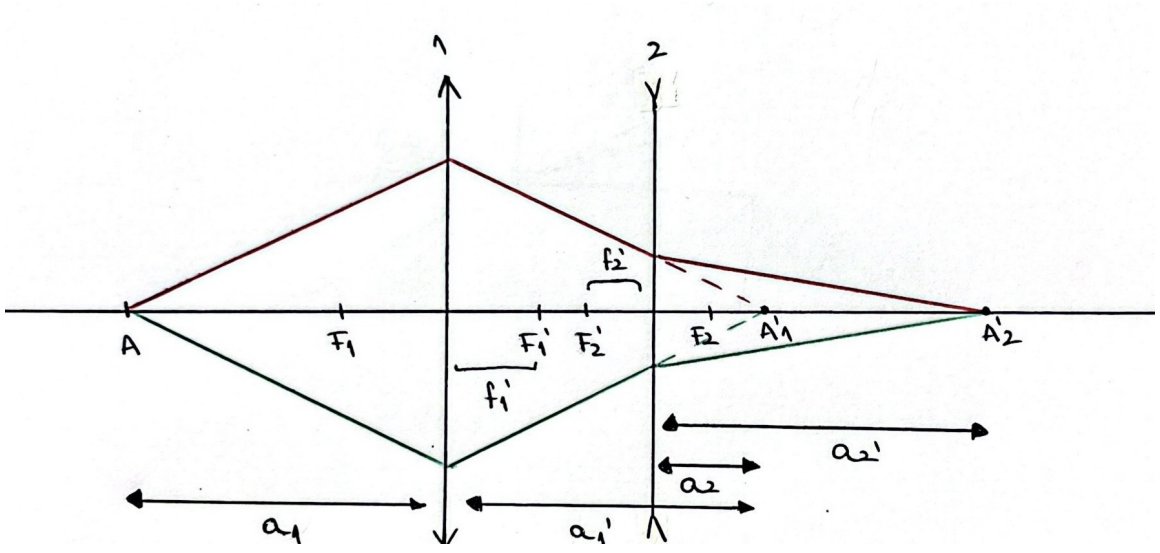



Figure 5: Situation described in which the image of the convex lens works as an object for the concave lens.

To locate this real image, the screen must be moved along the optical bench until a sharp focus is achieved. To obtain $a_2'$ we will measure the distance between the position of the image projected on the screen ($x_{I_2}$) and the concave lens ($x_{L_2}$). On the other hand, to obtain $a_2$, we will follow the next steps:

1. Determine the distance ($a_1$) between the original object (postioned in $x_{O_1}$) and the converging lens ($x_{L_1}$).

2. Using equation 6 we are able to calculate $a'_1$, which is the distance between the convex lens and the image it forms.

3. We obtain the position of the image, which is the new object, as follows:

$$x_{I_1} = x_{L_1} + a'_1(m) \tag{4}$$

4. Finally, $a_2$ is calculated with the relation 5

$$a_2 = x_{I_1} - x_{L_2}(m) \tag{5}$$

Once $a_2$ and $a_2$ are obtained, we will use equation 2 to determine the focal length ($f'_2$) of the concave lens, and equation 3 will be applied to calculate its optical power.

As in the previous phase, a ray-tracing diagram will be drawn to verify the accuracy of the method, using the theoretical value of $f'_1$.To conclude, it must be confirmed that the calculated values of $a_2$, $a'_2$, and $f'_2$ satisfy equation 2.

# 3 Results and discussion

## 3.1 Convex lens

The relevant data obtained experimentally is shown in the following table.

| $x_I$ (m) | Lens | $a_1$ (m) | $a'_1$ (m) | $a_2$ (m) | $a'_2$ (m) |
|---|---|---|---|---|---|
| 0,985 | 1 | $-0,78700 \pm 0,00041$ | $0,11400 \pm 0,00041$ | - | - |
| 0,985 | 2 | $-0,19900 \pm 0,00041$ | $0,70200 \pm 0,00041$ | $-0,71400 \pm 0,00041$ | $0,18700 \pm 0,00041$ |
| 0,985 | 3 | $-0,29600 \pm 0,00041$ | $0,60500 \pm 0,00041$ | $-0,60100 \pm 0,00041$ | 0,030 |
| 0,090 | 1 | $-0,69900 \pm 0,00041$ | $0,11700 \pm 0,00041$ | - | - |
| 0,090 | 2 | $-0,20700 \pm 0,00041$ | $0,60900 \pm 0,00041$ | $-0,61600 \pm 0,00041$ | $0,02000 \pm 0,00041$ |
| 0,090 | 3 | $-0,34100 \pm 0,00041$ | $0,47500 \pm 0,00041$ | $-0,49800 \pm 0,00041$ | $0,31800 \pm 0,00041$ |
| 0,085 | 1 | $-0,64700 \pm 0,00041$ | $0,11900 \pm 0,00041$ | - | - |
| 0,085 | 2 | $-0,21500 \pm 0,00041$ | $0,55100 \pm 0,00041$ | $-0,55800 \pm 0,00041$ | $0,20800 \pm 0,00041$ |
| 0,085 | 3 | $-0,03800 \pm 0,00041$ | $0,38600 \pm 0,00041$ | - | - |

Table 1: Position of screen $x_I$; distances $a$ and $a'$ taken from data sets 1 and 2; and lenses 1, 2 and 3, corresponding to the lens with a lower focal length, the medium and the larger one respectively.

With that, applying equation 2, focal lenght $f\prime$ has been found for each lens. That way, we can determinate:

| Lens | $f'_1$ (m) | $f'_2$ (m) |
|---|---|---|
| 1 | $0,09958 \pm 0,00032$ | - |
| | $0,10022 \pm 0,00030$ | - |
| | $0,10051 \pm 0,00030$ | - |
| 2 | $0,15505 \pm 0,00025$ | $0,148189 \pm 0,00026$ |
| | $0,15449 \pm 0,00023$ | $0,15098 \pm 0,00023$ |
| | $0,15465 \pm 0,00021$ | $0,15152 \pm 0,00022$ |
| 3 | $0,19876 \pm 0,00019$ | $0,20011 \pm 0,00019$ |
| | $0,19849 \pm 0,00016$ | $0,19407 \pm 0,00016$ |
| | $0,19149 \pm 0,00014$ | - |

Table 2: Calculated $f'_1$ and $f'_2$ from data sets 1 ($a_1$, $a'_1$) and 2 ($a_2$, $a'_2$)

Finally, from calculating the average of these found focal lengths, we can determine the frequency defined in equation 3.

| Lens | $\overline{f'}(m)$ | $\overline{\psi}$ (D) |
|---|---|---|
| 1 | $0,10010 \pm 0,00017$ | $0,099 \pm 0,017$ |
| 2 | $0,15248 \pm 0,00013$ | $0,0656 \pm 0,0057$ |
| 3 | $0,196586 \pm 0,000095$ | $0,0509 \pm 0,0025$ |

In some cases of this experiment, we can observe that a well-defined image emerges twice, at different distances. This happens because the image formed by a converging lens is virtual and upright when the object is placed between the lens and its focal length; and real and inverted when it is located at a distance between the focal length and twice that value, as well as when it is beyond twice the focal length. When the object is positioned exactly at a distance of $2f'$ from the converging lens, something quite interesting occurs: the image is real, inverted, and the same size as the object.

Therefore, in the trials where the image is formed twice, it is because we can place the lens at a distance within the interval $[f', 2f']$, and also at a distance greater than $2f'$.

Finally, we can generate with a simulator [3] a ray diagram of an example with real experimental measures.

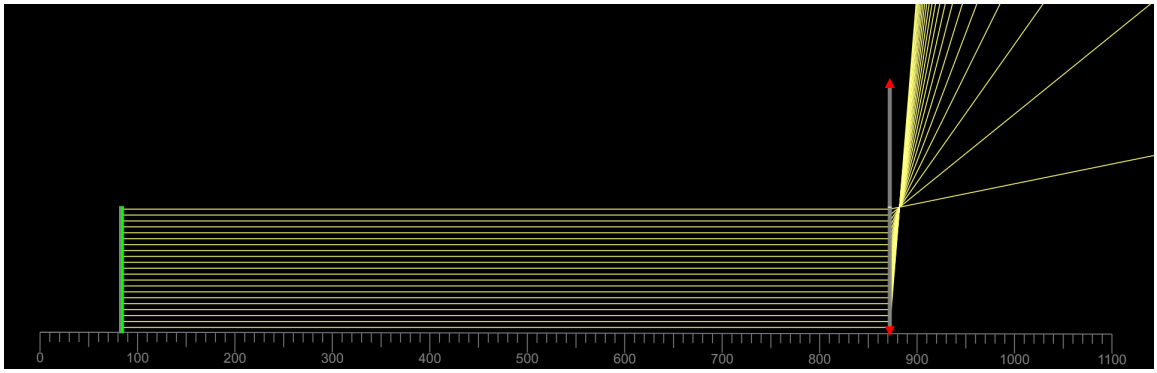



Figure 6: Ray diagram of a light refracting through convex lens whose focal length is 100 mm.

It is important to note that, as in our experiment, the screen where the object was found is, approximately, 780 mm far from the object. As the lens' focal length used in the simulator is 100 mm, and that is the value we have found with our deductions, this part of the experiments corresponds with reality properly.

### 3.2 Concave lens

In the first place, we need to calculate the distance $a'_1$ from the convex lens where the image forms. This image will be the object of the concave lens. Clearing $a'_1$ from the relation 1:

$$a'_1 = \frac{a_1 \cdot f'}{a_1 + f'}(m) \tag{6}$$

With this expression and the measurements of $a_1$ taken for each convergent lens, we constructed the following table.

| $f'(m)$ | $x_{O_1}(m)$ | $x_{L_1}(m)$ | $a_1(m)$ | $a'_1(m)$ |
|---|---|---|---|---|
| $0,100 \pm 0,001$ | $0,08400 \pm 0,00029$ | $0,84800 \pm 0,00029$ | $-0,76400 \pm 0,00041$ | $0,1151 \pm 0,0012$ |
| $0,150 \pm 0,001$ | $0,08400 \pm 0,00029$ | $0,28380 \pm 0,00029$ | $-0,19980 \pm 0,00041$ | $0,6018 \pm 0,0042$ |
| $0,200 \pm 0,001$ | $0,08400 \pm 0,00029$ | $0,41450 \pm 0,00029$ | $-0,33050 \pm 0,00041$ | $0,5065 \pm 0,0026$ |

Table 3: Focal lengths $f'$ and positions $x_{L_1}$ of each lens; position of the object 1 $x_{O_1}$; and distances $a_1$ and $a'_1$ that refer to the object and image length when the ligth goes through the converging lens.

The image formed by the converging is the lens is taken by the diverging one as the object. Using equations 4 and 5 we can obtain the position of the image formed by the convex lens, as well as the object distance $a_2$. The results of this part of the experiment are organised in table.

| $f'(m)$ | $x_{O_2}(m)$ | $x_{I_2}(m)$ | $x_{L_2}(m)$ | $a_2(m)$ | $a'_2(m)$ |
|---|---|---|---|---|---|
| $0,100 \pm 0,001$ | $0,9631 \pm 0,0012$ | $0,985 \pm 0,00029$ | $0,92400 \pm 0,00029$ | $0,03906 \pm 0,00041$ | $0,06100 \pm 0,00041$ |
| $0,150 \pm 0,001$ | $0,8856 \pm 0,0042$ | $0,985 \pm 0,00029$ | $0,81900 \pm 0,00029$ | $0,06661 \pm 0,00041$ | $0,16600 \pm 0,00041$ |
| $0,200 \pm 0,001$ | $0,9210 \pm 0,0026$ | $0,985 \pm 0,00029$ | $0,86700 \pm 0,00029$ | $0,05401 \pm 0,00041$ | $0,11800 \pm 0,00041$ |

Table 4: Focal lengths $f'$ of each convex lens; position of the object 2 $x_{O_2}$, which is the same as the $x_{I_1}$; position $x_{I_2}$ of the screen where the image originates, and $x_{L_2}$, where the diverging lens is positioned; and distances $a_2$ and $a'_2$ that refer to the object and image length when the ligth goes through the diverging lens.

Finally, using this data we can obtain the focal length of the diverging lens, as well as its power.

| $f'_1(m)$ | $f'_2(m)$ | $\psi_2(D)$ | $\overline{f'_2}(m)$ | $\overline{\psi}_2(D)$ |
|---|---|---|---|---|
| $0,100 \pm 0,001$ | $-0,1086 \pm 0,0034$ | $-0,09208 \pm 0,00097$ | | |
| $0,150 \pm 0,001$ | $-0,1112 \pm 0,0070$ | $-0,8989 \pm 0,0040$ | $-0,1065 \pm 0,0031$ | $-0,9412 \pm 0,0014$ |
| $0,200 \pm 0,001$ | $-0,0996 \pm 0,0049$ | $-0,100393 \pm 0,000024$ | | |

Table 5: Focal lengths of the convex $f'_1$ and concave lens $f'_2$; power $\psi_2$ of the diverging lens; and averages values $\overline{f'_2}$ and $\overline{\psi}_2$

The figure 7 is a ray diagram of a real case we measured experimentally, plotted with the simulator [3].

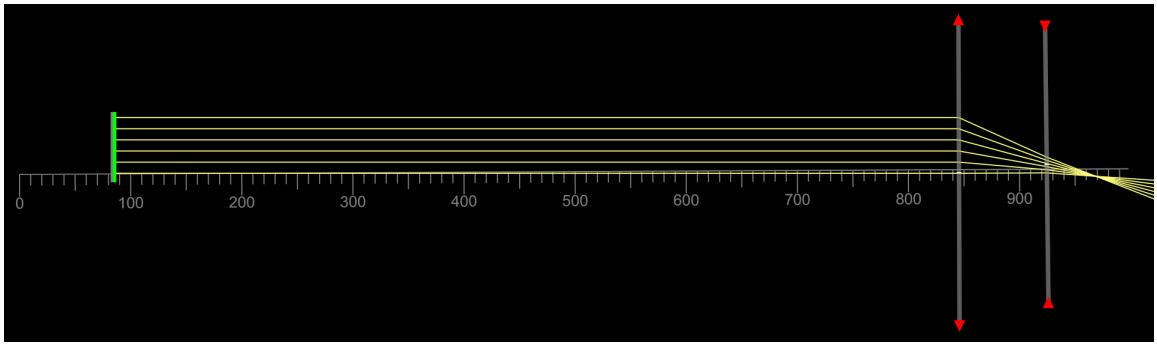



Figure 7: Ray diagram of a real situation where the light refractes in the convex lens and the image it forms is taken by the concave lens as the object.

As we can see in the diagram, the image originates at 98 cm, which is approximately where the screen is positioned in our experiment.

# 4 Conclusions

The results of the experiment successfully aligned with the theoretical expectations for both convex and concave lenses. The measured focal lengths of the convex lenses ($0,10010 \pm 0,00017$ mm for the first lens, $0,15248 \pm 0,00013$ mm for the second one and $0,196586 \pm 0,000095$ mm for the third) were consistent with their theoretical values ($0,100 \pm 0,001$, $0,150 \pm 0,001$ and $0,200 \pm 0,001$ respectively), with small uncertainties, confirming the validity of the experimental setup and methodology.

Analogously, the concave lens' focal length and optical power were accurately determined by using the real image produced by the convex lens as its object, obtaining a value for $f'$ of $-0,1065 \pm 0,0031$, very close to the number given by the manufacter ($-0,100 \pm 0,001$).

The uncertainty analysis confirmed that the measurements were reliable, and the observed discrepancies were within the expected error margins. Therefore, this experiment serves as a practical demonstration of lens behavior and image formation.

# 5 Appendixes

## 5.1 Calculation of Uncertainties

As we know, type A uncertainty, $u_A(x)$, which is associated to human error, can be calculated with the following equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}} \tag{7}$$

where $\sigma(x)$ represents the standard deviation and $N$ the number of observations.

On the other hand we get type B uncertainty, $u_B(x)$, related to the measuring instrument error from:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}} \tag{8}$$

where $\delta_x$ is the resolution of the measuring instrument ($\delta_{pos}$=0,001 m)

From them, $u_C(x)$ is expressed as:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)} \tag{9}$$

Additionally, we can calculate the expanded uncertainty is calculated to overestimate the error. To do so, we multiplied $u_C(x)$ by a coverage factor $kp$ that is selected for convenience:

$$U_C(x) = u_C(x)k_p \tag{10}$$

So, for all the positions, we get:

$$u_C(pos) = \sqrt{u_A^2(pos) + u_B^2(pos)} = \sqrt{0 + 0,00029^2} = 0,00029(m) \tag{11}$$

Regarding the error of the calculated distances, since they are indirect measurements obtained as the subtraction of positions, for a distance calculated as $a = pos_2 - pos_1$, we epxress:

$$u_C(a) = \sqrt{\left(\frac{\partial a}{\partial pos_1}\right)^2 (u_C(pos_1))^2 + \left(\frac{\partial a}{\partial pos_2}\right)^2 (u_C(pos_2))^2} \tag{12}$$

$$= \sqrt{0,00029^2 + 0,00029^2} = 0,00041(m)$$

To continue, $f'$, obtained indirectly from equation 2, gets an uncertainty $u_C(f')$:

$$u_C(f') = \sqrt{\left(\frac{\partial f'}{\partial a}\right)^2 (u_C(a))^2 + \left(\frac{\partial f'}{\partial a'}\right)^2 (u_C(a'))^2} \tag{13}$$

$$= \sqrt{\left(\frac{a'^2}{(a-a')^2}\right)^2 (u_C(a))^2 + \left(\frac{a^2}{(a-a')^2}\right)^2 (u_C(a'))^2}$$

To conclude the calculation of uncertainties of the first part, the error of the power $\psi$ is calculated, $u_C(\psi)$:

$$u_C(\psi) = \sqrt{\left(\frac{\partial \psi}{\partial f'}\right)^2 (u_C(f'))^2} = \frac{1}{f'^2}(u_C(f'))(D) \tag{14}$$

Beginning with uncertainties from the second part of the study, distance $a'_1$, an indirect measurement obtained with the formula 6, takes an error of:

$$u_C(a'_1) = \sqrt{\left(\frac{\partial a'_1}{\partial a_1}\right)^2 (u_C(a_1))^2 + \left(\frac{\partial a'_1}{\partial f'}\right)^2 (u_C(f'))^2} \tag{15}$$

$$= \sqrt{\left(\frac{f'^2}{(a_1+f')^2}\right)^2 (u_C(a_1))^2 + \left(\frac{a_1^2}{(a_1+f')^2}\right)^2 (u_C(f'))^2}(m)$$

Lastly, for distance $x_{I_1}$ and length $a_2$, also indirect measurements (4, 5), we express:

$$u_C(x_{I_1}) = \sqrt{\left(\frac{\partial x_{I_1}}{\partial x_{L_1}}\right)^2 (u_C(x_{L_1}))^2 + \left(\frac{\partial x_{I_1}}{\partial a'_1}\right)^2 (u_C(a'_1))^2} = \sqrt{(u_C(x_{L_1}))^2 + (u_C(a'_1))^2}(m) \tag{16}$$

$$u_C(a_2) = \sqrt{\left(\frac{\partial a_2}{\partial x_{I_1}}\right)^2 (u_C(x_{I_1}))^2 + \left(\frac{\partial a_2}{\partial x_{L_2}}\right)^2 (u_C(x_{L_2}))^2} = \sqrt{(u_C(x_{I_1}))^2 + (-u_C(x_{L_2}))^2}(m) \tag{17}$$

## References


[1] S.A. Akhmanov; S. Yu. Nikitin. *Physical optics*. Oxford University Press, 1997.

[2] I. Adroher; D. Bianco; A. Cazoría; J.R. Morilla; F. Navas; R. Rica; S. Koldan; G. Titos; A. Valenzuela. *Laboratory Manual for Basic Experimental Techniques*. Department of Applied Physics, University of Granada, 2025.

[3] *Simulador de trazado de rayos*. 2025. URL: https://phydemo.app/ray-optics/es/.